

Erdős Problem #342

EP-342

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

With $a_1 = 1$ and $a_2 = 2$ let a_{n+1} for $n \geq 2$ be the least integer $> a_n$ which can be expressed uniquely as $a_i + a_j$ for $i < j \leq n$.

What can be said about this sequence? Do infinitely many pairs $a, a + 2$ occur? Does this sequence eventually have periodic differences? Is the density 0?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>